Week 5 – Practice: Topic Modeling

Objective: Apply and compare topic modeling techniques, investigate the influence of document representation, and evaluate the coherence and interpretability of the extracted topics.

1. Dataset preparation

- Select three distinct subject areas (topics) and automatically retrieve 10 Wikipedia articles for each area using a Python library (e.g., wikipedia-api, wikipedia). Store the article titles, texts, and corresponding subject areas. Extract the textual content of each article.
- Apply text preprocessing: lowercasing, tokenization, lemmatization, stop-word removal.
- Construct two document-term matrices:
    - Bag-of-Words (CountVectorizer from scikit-learn)
    - TF-IDF (TfidfVectorizer from scikit-learn)

2. Topic modeling

Apply the following methods using Python and scikit-learn on each of the two document-term matrices:

1. Latent Semantic Analysis (TruncatedSVD)
2. Non-negative Matrix Factorization (NMF)
3. Latent Dirichlet Allocation (LatentDirichletAllocation)

For each model:

- Start with k=**3 topics**.
- Extract and display the **top 10 words** associated with each topic.
- Assign each document to its most representative topic. For LSA, use the topic/component with the largest absolute score.
- Examine whether documents from the same subject area are grouped together.
- Make scatter plots for representing words and documents using the obtained embeddings

3. Topic coherence evaluation

Investigate the topic coherence metrics available in the Gensim library.

- Compute metrics for each model.
- Compare the coherence scores and discuss whether they agree with your qualitative interpretation of the topics.
- Repeat the experiments for **5 topics** and analyze the effect of changing the number of topics.

4. Comparative analysis

Summarize your findings and answer the following questions:

1. How does the choice between Bag-of-Words and TF-IDF influence the topics extracted by LSA, NMF, LDA?
2. Which method produces the most interpretable topics?
3. Does a higher (Cv) score always correspond to more meaningful or useful topics?
4. How does increasing the number of topics affect topic coherence and topic redundancy?
5. Do the extracted topics correspond to the three original subject areas? Explain any differences.

Deliverable: A Jupyter Notebook containing also a brief discussion of the findings.

Libraries: scikit-learn, Gensim, and standard Python text preprocessing libraries.